

Supplementary Material

Supplementary Text S1

DNAShapeR Usage

DNAShapeR is a tool for predicting DNAShape data developed in R for ultra-fast, high-throughput prediction of DNA shape features. The package allows prediction, visualization and encoding of DNA shape features for statistical learning.

If you want to use it, you must first start R (version "4.4") and enter:

```
if (!require("BiocManager", quietly = TRUE))
```

```
    install.packages("BiocManager")
```

```
BiocManager::install("DNAShapeR")
```

However, its input must be in fasta file format, and the sequence files of the 165 datasets are in csv file format, so they need to be reformatted. The results are then predicted by the toolkit and then the results are saved.

Algorithm: Predicting DNA shape data

Input: DNA sequence data

Output: Shape data prediction files for five DNA shapes EP, HelT, MGW, ProT, Roll

1. **for** folder in all_folder:
 2. **read** csv file to data
 3. **Create** a vector of strings in fasta format fasta-sequences
 4. **for** row in data:
 5. **put** each row of data into fasta format into fasta-sequences
 6. **write** fasta data to fasta file
 7. **use** the getShape method to pass in a fasta file for prediction
 8. **save** results to file
 9. **end for**
-
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Supplementary Text S2

Validation of semantic alignment for the MSFFP module

We employed two quantitative metrics to assess the alignment between DNA sequence features and shape features:

- Mean Cosine Similarity, which reflects the directional consistency between token-level features across modalities. Higher values indicate stronger semantic alignment.
- L2 Distance, which measures the Euclidean distance between modality-specific feature centroids. A decrease in L2 distance suggests that the modalities are drawn closer in feature space, indicating better alignment.

We used the test set of the CTCF transcription factor dataset in the K562 cell line, consistent with our main experiments. Specifically, we computed the above metrics before applying the MSFFP module (i.e., on the original sequence and shape features), and again after the MSFFP fusion process.

Although the MSFFP module produces fused features, it does not disrupt channel ordering. Therefore, we were able to split the fused output by channel dimension to isolate the processed sequence and shape features, enabling direct comparison with their original forms.

Table 1: Semantic Alignment Verification Results

	Before MSFFP module	After MSFFP module
Mean Cosine Similarity	0.2769	0.3570
L2 Distance	63.8806	3.0375

As shown in Table 1, after applying the MSFFP module:

- Mean Cosine Similarity increased, indicating improved token-level directional alignment.
- L2 Distance significantly decreased, suggesting that the feature distributions of the two modalities were brought closer together.

These results provide quantitative evidence that the MSFFP module effectively enhances semantic alignment between DNA sequence and shape modalities.

Supplementary Text S3

contribution of DNA shape features

For each type of DNA shape, we trained a model using only that shape feature (while keeping other settings unchanged), and recorded its ACC, ROC-AUC, and PR-AUC scores.

To compute an overall contribution score for each DNA shape type, we applied a joint weighted formula:

$$\text{shape contribution} = 0.8 * ACC_{\text{shape}} + 0.2 * ROC - AUC_{\text{shape}} + 0.2 * PR - AUC_{\text{shape}}$$

This weighted strategy places more emphasis on classification accuracy while still taking into account the model's ranking and precision-recall performance. Compared to previous works (including the TBCA baseline), which typically rely on a single metric to assess contribution, our method provides a more comprehensive and balanced evaluation of each DNA shape feature's role in the final performance.

Supplementary Text S4

The impact of various fusion scales

We investigated the impact of using different scale divisions for feature fusion within the MSFFP module. The model was trained on the CTCF transcription factor dataset from the K562 cell line, with all other settings kept unchanged except for the scale configurations.

Scales size	ACC	ROC-AUC	PR-AUC
64	0.922	0.966	0.964
128	0.944	0.973	0.974
254	0.956	0.989	0.989

The smaller the scale, the more segments each feature map is divided into during fusion, which increases the complexity of the MSFFP module and may lead to overfitting. Through extensive hyperparameter tuning, we found that a scale size of 256 achieves the best balance between model performance and generalization.

Supplementary Table S1

Table S1. Details of 165 ChIP-seq datasets, including the cell line to which it belongs, the TF line, the number of positive and negative samples, and source organism information.

Cell lines	TFs	Datasets	Train_dataset		Test_dataset		Source organism
			Number of positive samples	Number of negative samples	Number of positive samples	Number of negative samples	
HepG2	CTCF	wgEncodeAwgTfbsBroadHepg2CtcfUniPk	72324		18082		Human Hepatocellular Carcinoma Cells
			36196	36128	9007	9075	
	EZH2	wgEncodeAwgTfbsBroadHepg2Ezh239875UniPk	5235		1309		
			2618	2617	654	655	
	CEBPB	wgEncodeAwgTfbsHaibHepg2Cebpbsc150V0416101UniPk	28491		7123		
			14243	14248	3564	3559	
	ELF1	wgEncodeAwgTfbsHaibHepg2Elf1sc631V0416101UniPk	28315		7079		
			14232	14083	3465	3614	
	FOS	wgEncodeAwgTfbsHaibHepg2Fosl2V0416101UniPk	39678		9920		
			19899	19779	4900	5020	
	FOXA1	wgEncodeAwgTfbsHaibHepg2Foxa1sc101058V0416101UniPk	67550		16888		
			33729	33821	8490	8398	
	FOXA1	wgEncodeAwgTfbsHaibHepg2Foxa1sc6553V0416101UniPk	80224		20056		
			40132	40092	10008	10048	
	GABP	wgEncodeAwgTfbsHaibHepg2GabpPcr2xUniPk	15710		3928		
			7871	7839	1948	1980	
	HDAC	wgEncodeAwgTfbsHaibHepg2Hdac2sc6296V0416101UniPk	29865		7467		
			14926	14939	3740	3727	
	JUND	wgEncodeAwgTfbsHaibHepg2JundPcr1xUniPk	34142		8536		

		17095	17047	4244	4292
P300	wgEncodeAwgTfbsHaibHepg2P300V0416101UniPk	43721		10931	
		21887	21834	5439	5492
SIN3A	wgEncodeAwgTfbsHaibHepg2Sin3ak20Pcr1xUniPk	26051		6513	
		13007	13044	3275	3238
SP1	wgEncodeAwgTfbsHaibHepg2Sp1Pcr1xUniPk	39952		9988	
		19891	20061	5079	4909
SP2	wgEncodeAwgTfbsHaibHepg2Sp2V0422111UniPk	4152		1038	
		2085	2067	510	528
SRF	wgEncodeAwgTfbsHaibHepg2SrfV0416101UniPk	8372		2094	
		4203	4169	1030	1064
TAF1	wgEncodeAwgTfbsHaibHepg2Taf1Pcr2xUniPk	25369		6343	
		12702	12667	3154	3189
TEAD4	wgEncodeAwgTfbsHaibHepg2Tead4sc101184V0422111UniPk	23852		5964	
		11933	11919	2975	2989
YY1	wgEncodeAwgTfbsHaibHepg2Yy1sc281V0416101UniPk	27944		6986	
		14065	13879	3400	3586
CEBPB	wgEncodeAwgTfbsSydhHepg2CebpbForsklnUniPk	29012		7254	
		14496	14516	3638	3616
CEBPB	wgEncodeAwgTfbsSydhHepg2CebpbIggrabUniPk	89115		22279	
		44654	44461	11044	11235
CJUN	wgEncodeAwgTfbsSydhHepg2CjunIggrabUniPk	20036		5010	
		10001	10035	2522	2488
CORES T	wgEncodeAwgTfbsSydhHepg2Corestsc30189IggrabUniPk	8515		2129	
		4271	4244	1051	1078
JUND	wgEncodeAwgTfbsSydhHepg2JundIggrabUniPk	50878		12720	

		25448	25430	6351	6369	
		18774		4694		
MAX	wgEncodeAwgTfbsSydhHepg2MaxIggrabUniPk	9382	9392	2352	2342	
		21732		5434		
TBP	wgEncodeAwgTfbsSydhHepg2TbpIggrabUniPk	10857	10875	2726	2708	
		7012		1754		
CMYC	wgEncodeAwgTfbsUtaHepg2CmycUniPk	3509	3503	874	880	
		80417		20105		
CTCF	wgEncodeAwgTfbsBroadK562CtcfUniPk	40042	40375	10219	9886	
		2684		672		
EZH2	wgEncodeAwgTfbsBroadK562Ezh239875UniPk	1366	1318	311	361	
		8380		2096		
HDAC	wgEncodeAwgTfbsBroadK562Hdac2a300705aUniPk	4185	4195	1053	1043	
		1779		445		
HDAC	wgEncodeAwgTfbsBroadK562Hdac6a301341aUniPk	884	895	228	217	
		34747		8687		
CEBPB	wgEncodeAwgTfbsHaibK562Cebpbc150V0422111UniPk	17353	17394	4364	4323	Human Chronic
K562		43403		10851		Myelogenous
ELF1	wgEncodeAwgTfbsHaibK562Elf1sc631V0416102UniPk	21787	21616	5340	5511	Leukemia Cells
		17569		4393		
FOS	wgEncodeAwgTfbsHaibK562Fosl1sc183V0416101UniPk	8782	8787	2199	2194	
		22387		5597		
GABP	wgEncodeAwgTfbsHaibK562GabpV0416101UniPk	11191	11196	2801	2796	
		30081		7521		
GATA2	wgEncodeAwgTfbsHaibK562Gata2sc267Pcr1xUniPk	15080	15001	3721	3800	
		10539		2635		
HDAC	wgEncodeAwgTfbsHaibK562Hdac2sc6296V0416102UniPk					

		5288	5251	1299	1336
MAX	wgEncodeAwgTfbsHaibK562MaxV0416102UniPk	70233		17559	
		34919	35314	8977	8582
SIN3A	wgEncodeAwgTfbsHaibK562Sin3ak20V0416101UniPk	19993		4999	
		9982	10011	2514	2485
SRF	wgEncodeAwgTfbsHaibK562SrfV0416101UniPk	7414		1854	
		3725	3689	909	945
TAF1	wgEncodeAwgTfbsHaibK562Taf1V0416101UniPk	23428		5858	
		11726	11702	2917	2941
TEAD4	wgEncodeAwgTfbsHaibK562Tead4sc101184V0422111UniPk	48372		12094	
		24181	24191	6052	6042
YY1	wgEncodeAwgTfbsHaibK562Yy1V0416101UniPk	19684		492	
		9882	9802	2421	2501
YY1	wgEncodeAwgTfbsHaibK562Yy1V0416102UniPk	37296		9324	
		18598	18698	4712	4612
CEBPB	wgEncodeAwgTfbsSydhK562CebpbIggrabUniPk	60416		15104	
		30270	30146	7492	7612
FOS	wgEncodeAwgTfbsSydhK562CfosUniPk	11968		2992	
		5985	5983	1495	1497
CJUN	wgEncodeAwgTfbsSydhK562CjunUniPk	15460		3866	
		7788	7672	1875	1991
COREST	wgEncodeAwgTfbsSydhK562Corestsc30189IggrabUniPk	56211		14053	
		28109	28102	7023	7030
GATA2	wgEncodeAwgTfbsSydhK562Gata2UcdUniPk	16649		4163	
		8323	8326	2083	2080
JUND	wgEncodeAwgTfbsSydhK562JundIggrabUniPk	62753		15689	

		31391	31362	7830	7859	
		49299		12325		
MAX	wgEncodeAwgTfbsSydhK562MaxIggrabUniPk	24653	24646	6159	6166	
		40425		10107		
P300	wgEncodeAwgTfbsSydhK562P300IggrabUniPk	20236	20189	5030	5077	
		27345		6837		
TBP	wgEncodeAwgTfbsSydhK562TbpIggmusUniPk	13654	13691	3437	3400	
		7638		1910		
YY1	wgEncodeAwgTfbsSydhK562Yy1UcdUniPk	3812	3826	962	948	
		16120		4030		
FOS	wgEncodeAwgTfbsUchicagoK562EfosUniPk	8064	8056	2011	2019	
		18004		4502		
GATA2	wgEncodeAwgTfbsUchicagoK562Egata2UniPk	9008	8996	2245	2257	
		2734		684		
HDAC	wgEncodeAwgTfbsUchicagoK562Ehdac8UniPk	1376	1358	333	351	
		41771		10443		
JUND	wgEncodeAwgTfbsUchicagoK562EjundUniPk	20954	20817	5153	5290	
		86808		21702		
CTCF	wgEncodeAwgTfbsUtaK562CtcfUniPk	43374	43434	10881	10821	
		13468		3368		
ELF1	wgEncodeAwgTfbsHaibA549Elf1V0422111Etoh02UniPk	6782	6686	1636	1732	
		44750		11188		
FOS	wgEncodeAwgTfbsHaibA549Fosl2V0422111Etoh02UniPk	22355	22395	5614	5574	Human Lung
A549		12054		3014		Adenocarcinoma Cells
FOXA1	wgEncodeAwgTfbsHaibA549Foxa1V0416102Dex100nmUniPk	6009	6045	1525	1489	
		18315		4579		
GABP	wgEncodeAwgTfbsHaibA549GabpV0422111Etoh02UniPk					

		9203	9112	2244	2335	
		9516		2380		
SIN3A	wgEncodeAwgTfbsHaibA549Sin3ak20V0422111Etoh02UniPk	4765	4751	1183	1197	
		15691		3923		
TAF1	wgEncodeAwgTfbsHaibA549Taf1V0422111Etoh02UniPk	7833	7858	1974	1949	
		60745		15187		
CEBPB	wgEncodeAwgTfbsSydhA549CebpbIggrabUniPk	30409	30336	7557	7630	
		103174		25794		
CTCF	wgEncodeAwgTfbsBroadH1hescCtcfUniPk	51646	51528	12838	12956	
		1758		440		
FOS	wgEncodeAwgTfbsHaibH1hescFosl1sc183V0416102UniPk	875	883	224	216	
		8968		2242		
GABP	wgEncodeAwgTfbsHaibH1hescGabpPcr1xUniPk	4490	4478	1115	1127	
		8953		2239		
HDAC	wgEncodeAwgTfbsHaibH1hescHdac2sc6296V0416102UniPk	4472	4481	1124	1115	
		23648		5912		
SP1	wgEncodeAwgTfbsHaibH1hescSp1Pcr1xUniPk	11833	11815	2947	2965	Human
		3830		958		Embryonic
SP2	wgEncodeAwgTfbsHaibH1hescSp2V0422111UniPk	1915	1915	479	479	Stem Cells
		8009		2003		
SRF	wgEncodeAwgTfbsHaibH1hescSrfPcr1xUniPk	4008	4001	998	1005	
		31233		7809		
TEAD4	wgEncodeAwgTfbsHaibH1hescTead4sc101184V0422111UniPk	15632	15601	3889	3920	
		28672		7168		
YY1	wgEncodeAwgTfbsHaibH1hescYy1sc281V0416102UniPk	14357	14315	3563	3605	
		24472		6118		
CEBPB	wgEncodeAwgTfbsSydhH1hescCebpbIggrabUniPk					

		12295	12177	3001	3117
		11228		2808	
TBP	wgEncodeAwgTfbsSydhH1hesCctbp2UcdUniPk	5625	5603	1393	1415
		15019		3755	
JUND	wgEncodeAwgTfbsSydhH1hesCjundIggrabUniPk	7532	7487	1855	1900
		17553		4389	
MAX	wgEncodeAwgTfbsSydhH1hesCmaxUcdUniPk	8709	8844	2262	2127
		33697		8425	
SIN3A	wgEncodeAwgTfbsSydhH1hesCSin3anb6001263IggrabUniPk	16812	16885	4249	4176
		1939		485	
CMYC	wgEncodeAwgTfbsUtaH1hesCmycUniPk	971	968	241	244
		67601		16901	
CTCF	wgEncodeAwgTfbsUtaH1hesCctcfUniPk	33756	33845	8495	8406
		81049		20263	
CTCF	wgEncodeAwgTfbsBroadHela3CtcfUniPk	40493	40556	10163	10100
		2840		710	
EZH2	wgEncodeAwgTfbsBroadHela3Ezh239875UniPk	1404	1436	371	339
		4779		1195	
POL2B	wgEncodeAwgTfbsBroadHela3Pol2bUniPk	2385	2394	602	593
		10494		2624	
GABP	wgEncodeAwgTfbsHaibHela3GabpPcr1xUniPk	5231	5263	1328	1296
		94892		23724	
CEBPB	wgEncodeAwgTfbsSydhHela3CebpblggrabUniPk	47379	47513	11930	11794
		14494		3624	
FOS	wgEncodeAwgTfbsSydhHela3CfosUniPk	7251	7243	1808	1816
		34089		8523	
CJUN	wgEncodeAwgTfbsSydhHela3CjunIggrabUniPk				

Human Cervical
Carcinoma
Cells

		17057	17032	4249	4274	
		15990		3998		
HAE2F1	wgEncodeAwgTfbsSydhHela3Hae2f1UniPk	7993	7997	2001	1997	
		49201		12301		
JUND	wgEncodeAwgTfbsSydhHela3JundIggrabUniPk	24647	24554	6104	6197	
		46209		11553		
MAX	wgEncodeAwgTfbsSydhHela3MaxIggrabUniPk	23129	23080	5752	5801	
		40262		10066		
P300	wgEncodeAwgTfbsSydhHela3P300sc584sc584IggrabUniPk	20156	20106	5008	5058	
		21563		5391		
STAT3	wgEncodeAwgTfbsSydhHela3Stat3IggrabUniPk	10740	10823	2737	2654	
		90883		22721		
CTCF	wgEncodeAwgTfbsUtaHela3CtcfUniPk	45460	45423	11343	11378	
		109712		27428		Human Fetal
IMR90	CEBPB	54861	54851	13710	13718	Lung Fibroblast Cells
		77531		19383		
CTCF	wgEncodeAwgTfbsBroadDnd41CtcfUniPk	38783	38748	9674	9709	Human T-cell
Dnd41		2681		671		Leukemia Cells
EZH2	wgEncodeAwgTfbsBroadDnd41Ezh239875UniPk	1329	1352	348	323	
		69435		17359		
CTCF	wgEncodeAwgTfbsBroadGm12878CtcfUniPk	34753	34682	8644	8715	
		35684		8922		Human B-
ELF1	wgEncodeAwgTfbsHaibGm12878Elf1sc631V0416101UniPk	17857	17827	4446	4476	Lymphoblastoid Cells
		10211		2553		
GABP	wgEncodeAwgTfbsHaibGm12878GabpPcr2xUniPk	5106	5105	1276	1277	
P300	wgEncodeAwgTfbsHaibGm12878P300Pcr1xUniPk	8052		2014		

		4034	4018	999	1015
Pax5	wgEncodeAwgTfbsHaibGm12878Pax5c20Pcr1xUniPk	39841		9961	
		20004	19837	4897	5064
Pax5	wgEncodeAwgTfbsHaibGm12878Pax5n19Pcr1xUniPk	31012		7754	
		15487	15525	3896	3858
TAF1	wgEncodeAwgTfbsHaibGm12878Taf1Pcr1xUniPk	22305		5577	
		11154	11151	2787	2790
YY1	wgEncodeAwgTfbsHaibGm12878Yy1sc281Pcr1xUniPk	48142		12036	
		24010	24132	6079	5957
FOS	wgEncodeAwgTfbsSydhGm12878CfosUniPk	3488		872	
		1749	1739	431	441
JUND	wgEncodeAwgTfbsSydhGm12878JundUniPk	3892		974	
		1974	1918	459	515
MAX	wgEncodeAwgTfbsSydhGm12878MaxIggmusUniPk	19747		4937	
		9872	9875	2470	2467
NFKB	wgEncodeAwgTfbsSydhGm12878NfkbTnfaIggrabUniPk	26700		6676	
		13345	13355	3343	3333
P300	wgEncodeAwgTfbsSydhGm12878P300bUniPk	9697		2425	
		4880	4817	1181	1244
STAT3	wgEncodeAwgTfbsSydhGm12878Stat3IggmusUniPk	10185		2547	
		5118	5067	1248	1299
TBP	wgEncodeAwgTfbsSydhGm12878TbpIggmusUniPk	23278		5820	
		11684	11594	2865	2955
CMYC	wgEncodeAwgTfbsUtaGm12878CmycUniPk	5830		1458	
		2923	2907	721	737
CTCF	wgEncodeAwgTfbsUtaGm12878CtcfUniPk	75536		18884	

			37790	37746	9420	9464	
Hmec	CTCF	wgEncodeAwgTfbsBroadHmecCtcfUniPk	61801		15451		Human Mammary Epithelial Cells
			30852	30949	7774	7677	
	CTCF	wgEncodeAwgTfbsBroadHsmmCtcfUniPk	79036		19760		
			39592	39444	9806	9954	
Hsmm	EZH2	wgEncodeAwgTfbsBroadHsmmEzh239875UniPk	2464		616		Human Skeletal Muscle Myoblast Cells
			1239	1225	300	316	
	CTCF	wgEncodeAwgTfbsBroadHsmtCtcfUniPk	74404		18602		
			37327	37077	9176	9426	
	CTCF	wgEncodeAwgTfbsBroadHuvecCtcfUniPk	58102		14526		
			29058	29044	7256	7270	
	EZH2	wgEncodeAwgTfbsBroadHuvecEzh239875UniPk	6788		1698		
			3397	3391	845	853	
	POL2B	wgEncodeAwgTfbsBroadHuvecPol2bUniPk	8334		2084		
			4187	4147	1022	1062	
Huvec	FOS	wgEncodeAwgTfbsSydhHuvecCfosUcdUniPk	72961		18241		Human Umbilical Vein Endothelial Cells
			36461	36500	9140	9101	
	CJUN	wgEncodeAwgTfbsSydhHuvecCjunUniPk	46128		11532		
			23077	23051	5753	5779	
	GATA2	wgEncodeAwgTfbsSydhHuvecGata2UcdUniPk	42681		10671		
			21339	21342	5337	5334	
	MAX	wgEncodeAwgTfbsSydhHuvecMaxUniPk	14409		3603		
			7214	7195	1792	1811	
Nha	CTCF	wgEncodeAwgTfbsBroadNhaCtcfUniPk	59652		14914		Human Adrenal Cortex Cells
			29817	29835	7466	7448	
Nhdfad	CTCF	wgEncodeAwgTfbsBroadNhdfadCtcfUniPk	75851		18963		

				37984	37867	9423	9540	Human Dermal Fibroblast Adult
				72955		18239		
	CTCF	wgEncodeAwgTfbsBroadNhekCtcfUniPk		36566	36389	9031	9208	Human Renal Proximal
Nhek				9046		2262		Tubule
	POL2B	wgEncodeAwgTfbsBroadNhekPol2bUniPk		4516	4530	1138	1124	Epithelial Cells
				61337		15335		
Nhlf	CTCF	wgEncodeAwgTfbsBroadNhlfCtcfUniPk		30645	30692	7691	7644	Human Lung Fibroblast
				86148		21538		
Osteobl	CTCF	wgEncodeAwgTfbsBroadOsteoblCtcfUniPk		42935	43213	10908	10630	Human Osteoblast
				9283		2321		
Ecc1	FOXA1	wgEncodeAwgTfbsHaibEcc1Foxa1sc6553V0416102Dm002p1hUniPk		4671	4612	1131	1190	Human Endometrial Carcinoma Cells
				4979		1245		
	Pax5	wgEncodeAwgTfbsHaibGm12891Pax5c20V0416101UniPk		2497	2482	615	630	Human B-
Gm1289				45032		11258		Lymphocyte
1	NFKB	wgEncodeAwgTfbsSydhGm12891NfkbTnfalIggrabUniPk		22489	22543	5656	5602	
				16284		4072		
	PAX5	wgEncodeAwgTfbsHaibGm12892Pax5c20V0416101UniPk		8076	8208	2102	1970	
				14776		3694		
	TAF1	wgEncodeAwgTfbsHaibGm12892Taf1V0416102UniPk		7381	7395	1854	1840	Human B-
Gm1289				24763		6191		Lymphocyte
2	YY1	wgEncodeAwgTfbsHaibGm12892Yy1V0416101UniPk		12306	12457	3171	3020	
				12865		3217		
	NFKB	wgEncodeAwgTfbsSydhGm12892NfkbTnfalIggrabUniPk		6423	6442	1618	1599	
				20043		5011		
Hct116	YY1	wgEncodeAwgTfbsHaibHct116Yy1sc281V0416101UniPk		9998	10045	2529	2482	Human Colon Carcinoma Cells
				11289		2823		
Panc1	SIN3A	wgEncodeAwgTfbsHaibPanc1Sin3ak20V0416101UniPk						

					5690	5599	1366	1457	Human Pancreatic Carcinoma Cells
	P300	wgEncodeAwgTfbsHaibSknshraP300V0416102UniPk			73086		18272		
Sknshra					36487	36599	9192	9080	Human Neuroblastoma Cells
	YY1	wgEncodeAwgTfbsHaibSknshraYy1sc281V0416102UniPk			24422		6106		
					12189	12233	3075	3031	
	P300	wgEncodeAwgTfbsHaibT47dP300V0416102Dm002p1hUniPk			21969		5493		
					10977	10992	2754	2739	
T47d	FOXA1	wgEncodeAwgTfbsHaibT47dFoxa1sc6553V0416102Dm002p1hUniPk			65257		16315		Human Breast Carcinoma Cells
					32624	32633	8162	8153	
	GATA3	wgEncodeAwgTfbsHaibT47dGata3sc268V0416102Dm002p1hUniPk			58292		14574		
					29070	29222	7363	7211	
Gm10847	NFKB	wgEncodeAwgTfbsSydhGm10847NfkbTnfalgrabUniPk			14545		3637		Human Prostate Carcinoma Cells
					7234	7311	1857	1780	
	CMYC	wgEncodeAwgTfbsSydhNb4CmycUniPk			41251		10313		
Nb4					20572	20679	5210	5103	Human Acute Promyelocytic Leukemia Cells
	MAX	wgEncodeAwgTfbsSydhNb4MaxUniPk			54438		13610		
					27209	27229	6815	6795	
	Sknsh	TAF1	wgEncodeAwgTfbsHaibSknshTaf1V0416101UniPk		21908		5478		Human Neuroblastoma Cells
					11002	10906	2691	2787	
Gm18526	NFKB	wgEncodeAwgTfbsSydhGm18526NfkbTnfalgrabUniPk			4820		1206		Human Lymphoblast Cells
					2395	2425	617	589	
Gm19099	NFKB	wgEncodeAwgTfbsSydhGm19099NfkbTnfalgrabUniPk			12190		3048		Human Lymphoblast Cells
					6073	6117	1546	1502	
Mcf10aes	FOS	wgEncodeAwgTfbsSydhMcf10aesCfosTam112hHvdUniPk			143425		35857		Human Mammary Epithelial Cells
					71850	71575	17791	18066	

				134080	33520	
FOS		wgEncodeAwgTfbsSydhMcf10aesCfosTam14hHvdUniPk		67066	67014	16734 16786
				109454	27364	
FOS		wgEncodeAwgTfbsSydhMcf10aesCfosTamHvdUniPk		54707	54747	13702 13662
				70411	17603	
STAT3		wgEncodeAwgTfbsSydhMcf10aesStat3Etoh01bUniPk		35281	35130	8726 8877
				64587	16147	
STAT3		wgEncodeAwgTfbsSydhMcf10aesStat3Etoh01cUniPk		32324	32263	8043 8104
				19377	4845	
STAT3		wgEncodeAwgTfbsSydhMcf10aesStat3Etoh01UniPk		9713	9664	2398 2447
				62265	15567	
STAT3		wgEncodeAwgTfbsSydhMcf10aesStat3Tam112hHvdUniPk		31109	31156	7807 7760
				69246	17312	
STAT3		wgEncodeAwgTfbsSydhMcf10aesStat3TamUniPk		34662	34584	8617 8695
				7465	1867	Human Embryonic Carcinoma Cells
Nt2d1	YY1	wgEncodeAwgTfbsSydhNt2d1Yy1UcdUniPk		3690	3775	976 891
				55936	13984	
GATA2		wgEncodeAwgTfbsSydhShsy5yGata2UcdUniPk		27935	28001	7025 6959
Shsy5y				24948	6238	Human Neuroblastoma Cells
GATA3		wgEncodeAwgTfbsSydhShsy5yGata3sc269sc269UcdUniPk		12440	12508	3153 3085
				9585	2397	
GATA3		wgEncodeAwgTfbsSydhMcf7Gata3UcdUniPk		4793	4792	1198 1199
				30704	7676	Human Breast Carcinoma Cells
Mcf7	HAE2F1	wgEncodeAwgTfbsSydhMcf7Hae2f1UcdUniPk		15377	15327	3813 3863
				90049	22513	
CTCF		wgEncodeAwgTfbsUtaMcf7CtcfUniPk		45057	44992	11224 11289

Supplementary Table S2

Table S2. Partial Hyper-parameter settings of TEMNet.

parameters	Search space	optimal parameter
Epoch	15,40,60,100	40
Learning rate	0.01,0.001,0.0015,0.0018,0.002	0.0018
Batch size	32,64,128	64
Optimizer	Adam, AdamW, SGD	AdamW
dropout ratio	0.1,0.2,0.4,0.6	0.4
EMA factor	1,2,4	4
MLP hidden units	[512,192,1]	[512,192,1]

Supplementary Table S3

Table S3. The detailed settings of the sequence data section.

Architectures	Settings	Output shape
Bi-LSTM Layer	input_size=12 hidden_size=64 num_layers=2	(64,99,128)
Res1	input_channels=128 output channels=128 num_residuals=2	(64,128,1,99)
Res2	input_channels=128 output channels=256 num_residuals=2	(64,256,1,50)
Convolution_seq Layer1	Batch Norm-feature=256 ELU kernel number = 256 kernel size = (1, 3) stride = (1, 1)	(64,256,1,48)
Dropout Layer	dropout=0.4	(64,256,1,48)
Convolution_seq Layer2	Batch Norm-feature=256 ELU kernel number = 256 kernel size = (1, 3) stride = (1, 1)	(64,256,1,46)
Max-pooling Layer1	kernel size=(1, 3) stride = (1, 1)	(64,256,1,44)

Supplementary Table S4

Table S4. The detailed settings of the shape data section.

Architectures	Settings	Output shape
Conv_to_TA	input_channels=1 output channels=8 kernel size = (1, 1)	(64,8,5,101)
TA Layer	input_channels=8 output channels=64	(64,256,1,7)
Convolution_shape Layer1	kernel number = 64 kernel size = (1, 3) stride = (1, 1) ELU Batch Norm- feature=64	(64,256,1,7)
Max-pooling Layer1	kernel size=(1, 3) stride = (1, 1)	(64,256,1,5)
Dropout Layer	dropout=0.4	(64,256,1,5)
Bi-LSTM Layer	input_size=5 hidden_size=5 num_layers=16	(64,256,10)
Convolution_shape Layer2	Batch Norm- feature=64 ELU kernel number = 64 kernel size = (1, 3) stride = (1, 1)	(64,256,1,10)
Max-pooling Layer2	kernel size=(1, 3) stride = (1, 1)	(64,256,1,8)

Supplementary Table S5

Table S5. The detailed settings of the enhanced fusion feature extraction modules.

Architectures	Settings	Output shape
EMA	channels=256 factor=4	(64,256,1,52)
Convolution Layer1	kernel number = 256 kernel size = (1, 5) stride = (1, 1) ELU Batch Norm-feature=512	(64,256,1,48)
Max-pooling Layer1	kernel size=(1, 3) stride = (1, 1)	(64,256,1,46)

Supplementary Table S6

Table S6. The detailed settings of the enhanced fusion feature extraction

Models	Memory (MB)	Training time (ms)	Params (M)	FLOPs (G)	ROC-AUC
<i>DeepCTMS</i>	2454	1770	2.879	8.388	0.902
<i>MLSNet</i>	2472	1710	3.044	17.161	0.899
<i>DeepSTF</i>	1994	1250	3.706	2.042	0.882
<i>CRPTS</i>	1604	240	0.009	11.887	0.867
<i>D-SSCA</i>	1540	200	54.788	23.832	0.871
<i>DanQ</i>	1696	230	0.574	64.559	0.854

Supplementary Table S7

Table S7. Overlap of all transcription factors in 165 ChIP-seq datasets.

Cell lines	TF lines
Cells: HepG2 & K562	Shared transcription factors (18): CEBPB, COREST, CTCF, ELF1, EZH2, FOS, GABP, HDAC, CJUN, JUND, MAX, P300, SIN3A, SRF, TAF1, TBP, TEAD4, YY1
Cells: HepG2 & A549	Shared transcription factors (7): CEBPB, ELF1, FOS, FOXA1, GABP, SIN3A, TAF1
Cells: HepG2 & H1hesc	Shared transcription factors (15): CEBPB, CTCF, FOS, GABP, HDAC, CJUN, MAX, CMYC, SIN3A, SP1, SP2, SRF, TBP, TEAD4, YY1
Cells: HepG2 & HeLaS3	Shared transcription factors (9): CEBPB, CTCF, EZH2, FOS, GABP, CJUN, JUND, MAX, P300
Cells: HepG2 & IMR90	Shared transcription factors (1): CEBPB
Cells: HepG2 & Dnd41	Shared transcription factors (2): CTCF, EZH2
Cells: HepG2 & Gm12878	Shared transcription factors (11): CTCF, ELF1, FOS, GABP, JUND, MAX, CMYC, P300, TAF1, TBP, YY1
Cells: HepG2 & Hmec	Shared transcription factors (1): CTCF
Cells: HepG2 & Hsmm	Shared transcription factors (2): CTCF, EZH2
Cells: HepG2 & Huvec	Shared transcription factors (5): CTCF, EZH2, FOS, CJUN, MAX
Cells: HepG2 & Nha	Shared transcription factors (1): CTCF
Cells: HepG2 & Nhdfad	Shared transcription factors (1): CTCF
Cells: HepG2 & NHEK	Shared transcription factors (1): CTCF
Cells: HepG2 & NHLF	Shared transcription factors (1): CTCF
Cells: HepG2 & Osteobl	Shared transcription factors (1): CTCF
Cells: HepG2 & Ecc1	Shared transcription factors (1): FOXA1
Cells: HepG2 & Gm12892	Shared transcription factors (2): TAF1, YY1
Cells: HepG2 & HCT116	Shared transcription factors (1): YY1
Cells: HepG2 & Panc1	Shared transcription factors (1): SIN3A
Cells: HepG2 & SkNSHRA	Shared transcription factors (2): P300, YY1
Cells: HepG2 & T47D	Shared transcription factors (2): FOXA1, P300
Cells: HepG2 & NB4	Shared transcription factors (2): MAX, CMYC
Cells: HepG2 & SkNSH	Shared transcription factors (1): TAF1
Cells: HepG2 & Mcf10aes	Shared transcription factors (1): FOS
Cells: HepG2 & NT2D1	Shared transcription factors (1): YY1
Cells: HepG2 & MCF7	Shared transcription factors (1): CTCF
Cells: K562 & A549	Shared transcription factors (6): CEBPB, ELF1, FOS, GABP, SIN3A, TAF1
Cells: K562 & H1hesc	Shared transcription factors (12): CEBPB, CTCF, FOS, GABP, HDAC, CJUN, MAX, SIN3A, SRF, TBP, TEAD4, YY1
Cells: K562 & HeLaS3	Shared transcription factors (9): CEBPB, CTCF, EZH2, FOS, GABP, CJUN, JUND, MAX, P300
Cells: K562 & IMR90	Shared transcription factors (1): CEBPB

Cells: K562 & Dnd41	Shared transcription factors (2): CTCF, EZH2
Cells: K562 & Gm12878	Shared transcription factors (10): CTCF, ELF1, FOS, GABP, JUND, MAX, P300, TAF1, TBP, YY1
Cells: K562 & Hmec	Shared transcription factors (1): CTCF
Cells: K562 & Hsmm	Shared transcription factors (2): CTCF, EZH2
Cells: K562 & Huvec	Shared transcription factors (6): CTCF, EZH2, FOS, GATA2, CJUN, MAX
Cells: K562 & Nha	Shared transcription factors (1): CTCF
Cells: K562 & Nhdfad	Shared transcription factors (1): CTCF
Cells: K562 & NHEK	Shared transcription factors (1): CTCF
Cells: K562 & NHLF	Shared transcription factors (1): CTCF
Cells: K562 & Osteobl	Shared transcription factors (1): CTCF
Cells: K562 & Gm12892	Shared transcription factors (2): TAF1, YY1
Cells: K562 & HCT116	Shared transcription factors (1): YY1
Cells: K562 & Panc1	Shared transcription factors (1): SIN3A
Cells: K562 & SkNSHRA	Shared transcription factors (2): P300, YY1
Cells: K562 & T47D	Shared transcription factors (1): P300
Cells: K562 & NB4	Shared transcription factors (1): MAX
Cells: K562 & SkNSH	Shared transcription factors (1): TAF1
Cells: K562 & Mcf10aes	Shared transcription factors (1): FOS
Cells: K562 & NT2D1	Shared transcription factors (1): YY1
Cells: K562 & Shsy5y	Shared transcription factors (1): GATA2
Cells: K562 & MCF7	Shared transcription factors (1): CTCF
Cells: A549 & H1hesc	Shared transcription factors (4): CEBPB, FOS, GABP, SIN3A
Cells: A549 & HeLaS3	Shared transcription factors (3): CEBPB, FOS, GABP
Cells: A549 & IMR90	Shared transcription factors (1): CEBPB
Cells: A549 & Gm12878	Shared transcription factors (4): ELF1, FOS, GABP, TAF1
Cells: A549 & Huvec	Shared transcription factors (1): FOS
Cells: A549 & Ecc1	Shared transcription factors (1): FOXA1
Cells: A549 & Gm12892	Shared transcription factors (1): TAF1
Cells: A549 & Panc1	Shared transcription factors (1): SIN3A
Cells: A549 & T47D	Shared transcription factors (1): FOXA1
Cells: A549 & SkNSH	Shared transcription factors (1): TAF1
Cells: A549 & Mcf10aes	Shared transcription factors (1): FOS
Cells: H1hesc & HeLaS3	Shared transcription factors (6): CEBPB, CTCF, FOS, GABP, CJUN, MAX
Cells: H1hesc & IMR90	Shared transcription factors (1): CEBPB
Cells: H1hesc & Dnd41	Shared transcription factors (1): CTCF
Cells: H1hesc & Gm12878	Shared transcription factors (7): CTCF, FOS, GABP, MAX, CMYC, TBP, YY1
Cells: H1hesc & Hmec	Shared transcription factors (1): CTCF
Cells: H1hesc & Hsmm	Shared transcription factors (1): CTCF
Cells: H1hesc & Huvec	Shared transcription factors (4): CTCF, FOS, CJUN, MAX
Cells: H1hesc & Nha	Shared transcription factors (1): CTCF
Cells: H1hesc & Nhdfad	Shared transcription factors (1): CTCF
Cells: H1hesc & NHEK	Shared transcription factors (1): CTCF

Cells: H1hesc & NHLF	Shared transcription factors (1): CTCF
Cells: H1hesc & Osteobl	Shared transcription factors (1): CTCF
Cells: H1hesc & Gm12892	Shared transcription factors (1): YY1
Cells: H1hesc & HCT116	Shared transcription factors (1): YY1
Cells: H1hesc & Panc1	Shared transcription factors (1): SIN3A
Cells: H1hesc & SkNSHRA	Shared transcription factors (1): YY1
Cells: H1hesc & NB4	Shared transcription factors (2): MAX, CMYC
Cells: H1hesc & Mcf10aes	Shared transcription factors (1): FOS
Cells: H1hesc & NT2D1	Shared transcription factors (1): YY1
Cells: H1hesc & MCF7	Shared transcription factors (1): CTCF
Cells: HeLaS3 & IMR90	Shared transcription factors (1): CEBPB
Cells: HeLaS3 & Dnd41	Shared transcription factors (2): CTCF, EZH2
Cells: HeLaS3 & Gm12878	Shared transcription factors (7): CTCF, FOS, GABP, JUND, MAX, P300, STAT3
Cells: HeLaS3 & Hmec	Shared transcription factors (1): CTCF
Cells: HeLaS3 & Hsmm	Shared transcription factors (2): CTCF, EZH2
Cells: HeLaS3 & Huvec	Shared transcription factors (6): CTCF, EZH2, FOS, CJUN, MAX, POL2B
Cells: HeLaS3 & Nha	Shared transcription factors (1): CTCF
Cells: HeLaS3 & Nhdfad	Shared transcription factors (1): CTCF
Cells: HeLaS3 & NHEK	Shared transcription factors (2): CTCF, POL2B
Cells: HeLaS3 & NHLF	Shared transcription factors (1): CTCF
Cells: HeLaS3 & Osteobl	Shared transcription factors (1): CTCF
Cells: HeLaS3 & SkNSHRA	Shared transcription factors (1): P300
Cells: HeLaS3 & T47D	Shared transcription factors (1): P300
Cells: HeLaS3 & NB4	Shared transcription factors (1): MAX
Cells: HeLaS3 & Mcf10aes	Shared transcription factors (2): FOS, STAT3
Cells: HeLaS3 & MCF7	Shared transcription factors (2): CTCF, HAE2F1
Cells: Dnd41 & Gm12878	Shared transcription factors (1): CTCF
Cells: Dnd41 & Hmec	Shared transcription factors (1): CTCF
Cells: Dnd41 & Hsmm	Shared transcription factors (2): CTCF, EZH2
Cells: Dnd41 & Huvec	Shared transcription factors (2): CTCF, EZH2
Cells: Dnd41 & Nha	Shared transcription factors (1): CTCF
Cells: Dnd41 & Nhdfad	Shared transcription factors (1): CTCF
Cells: Dnd41 & NHEK	Shared transcription factors (1): CTCF
Cells: Dnd41 & NHLF	Shared transcription factors (1): CTCF
Cells: Dnd41 & Osteobl	Shared transcription factors (1): CTCF
Cells: Dnd41 & MCF7	Shared transcription factors (1): CTCF
Cells: Gm12878 & Hmec	Shared transcription factors (1): CTCF
Cells: Gm12878 & Hsmm	Shared transcription factors (1): CTCF
Cells: Gm12878 & Huvec	Shared transcription factors (3): CTCF, FOS, MAX
Cells: Gm12878 & Nha	Shared transcription factors (1): CTCF
Cells: Gm12878 & Nhdfad	Shared transcription factors (1): CTCF
Cells: Gm12878 & NHEK	Shared transcription factors (1): CTCF
Cells: Gm12878 & NHLF	Shared transcription factors (1): CTCF

Cells: Gm12878 & Osteobl	Shared transcription factors (1): CTCF
Cells: Gm12878 & Gm12891	Shared transcription factors (2): NFkB, PAX5
Cells: Gm12878 & Gm12892	Shared transcription factors (4): NFkB, PAX5, TAF1, YY1
Cells: Gm12878 & HCT116	Shared transcription factors (1): YY1
Cells: Gm12878 & SkNSHRA	Shared transcription factors (2): P300, YY1
Cells: Gm12878 & T47D	Shared transcription factors (1): P300
Cells: Gm12878 & Gm10847	Shared transcription factors (1): NFkB
Cells: Gm12878 & NB4	Shared transcription factors (2): MAX, CMYC
Cells: Gm12878 & SkNSH	Shared transcription factors (1): TAF1
Cells: Gm12878 & Gm18526	Shared transcription factors (1): NFkB
Cells: Gm12878 & Gm19099	Shared transcription factors (1): NFkB
Cells: Gm12878 & Mcf10aes	Shared transcription factors (2): FOS, STAT3
Cells: Gm12878 & NT2D1	Shared transcription factors (1): YY1
Cells: Gm12878 & MCF7	Shared transcription factors (1): CTCF
Cells: Hmec & Hsmm	Shared transcription factors (1): CTCF
Cells: Hmec & Huvec	Shared transcription factors (1): CTCF
Cells: Hmec & Nha	Shared transcription factors (1): CTCF
Cells: Hmec & Nhdaf	Shared transcription factors (1): CTCF
Cells: Hmec & NHEK	Shared transcription factors (1): CTCF
Cells: Hmec & NHLF	Shared transcription factors (1): CTCF
Cells: Hmec & Osteobl	Shared transcription factors (1): CTCF
Cells: Hmec & MCF7	Shared transcription factors (1): CTCF
Cells: Hsmm & Huvec	Shared transcription factors (2): CTCF, EZH2
Cells: Hsmm & Nha	Shared transcription factors (1): CTCF
Cells: Hsmm & Nhdaf	Shared transcription factors (1): CTCF
Cells: Hsmm & NHEK	Shared transcription factors (1): CTCF
Cells: Hsmm & NHLF	Shared transcription factors (1): CTCF
Cells: Hsmm & Osteobl	Shared transcription factors (1): CTCF
Cells: Hsmm & MCF7	Shared transcription factors (1): CTCF
Cells: Huvec & Nha	Shared transcription factors (1): CTCF
Cells: Huvec & Nhdaf	Shared transcription factors (1): CTCF
Cells: Huvec & NHEK	Shared transcription factors (2): CTCF, POL2B
Cells: Huvec & NHLF	Shared transcription factors (1): CTCF
Cells: Huvec & Osteobl	Shared transcription factors (1): CTCF
Cells: Huvec & NB4	Shared transcription factors (1): MAX
Cells: Huvec & Mcf10aes	Shared transcription factors (1): FOS
Cells: Huvec & Shsy5y	Shared transcription factors (1): GATA2
Cells: Huvec & MCF7	Shared transcription factors (1): CTCF
Cells: Nha & Nhdaf	Shared transcription factors (1): CTCF
Cells: Nha & NHEK	Shared transcription factors (1): CTCF
Cells: Nha & NHLF	Shared transcription factors (1): CTCF
Cells: Nha & Osteobl	Shared transcription factors (1): CTCF
Cells: Nha & MCF7	Shared transcription factors (1): CTCF

Cells: Nhdad & NHEK	Shared transcription factors (1): CTCF
Cells: Nhdad & NHLF	Shared transcription factors (1): CTCF
Cells: Nhdad & Osteobl	Shared transcription factors (1): CTCF
Cells: Nhdad & MCF7	Shared transcription factors (1): CTCF
Cells: NHEK & NHLF	Shared transcription factors (1): CTCF
Cells: NHEK & Osteobl	Shared transcription factors (1): CTCF
Cells: NHEK & MCF7	Shared transcription factors (1): CTCF
Cells: NHLF & Osteobl	Shared transcription factors (1): CTCF
Cells: NHLF & MCF7	Shared transcription factors (1): CTCF
Cells: Osteobl & MCF7	Shared transcription factors (1): CTCF
Cells: Ecc1 & T47D	Shared transcription factors (1): FOXA1
Cells: Gm12891 & Gm12892	Shared transcription factors (2): NFkB, PAX5
Cells: Gm12891 & Gm10847	Shared transcription factors (1): NFkB
Cells: Gm12891 & Gm18526	Shared transcription factors (1): NFkB
Cells: Gm12891 & Gm19099	Shared transcription factors (1): NFkB
Cells: Gm12892 & HCT116	Shared transcription factors (1): YY1
Cells: Gm12892 & SkNSHRA	Shared transcription factors (1): YY1
Cells: Gm12892 & Gm10847	Shared transcription factors (1): NFkB
Cells: Gm12892 & SkNSH	Shared transcription factors (1): TAF1
Cells: Gm12892 & Gm18526	Shared transcription factors (1): NFkB
Cells: Gm12892 & Gm19099	Shared transcription factors (1): NFkB
Cells: Gm12892 & NT2D1	Shared transcription factors (1): YY1
Cells: HCT116 & SkNSHRA	Shared transcription factors (1): YY1
Cells: HCT116 & NT2D1	Shared transcription factors (1): YY1
Cells: SkNSHRA & T47D	Shared transcription factors (1): P300
Cells: SkNSHRA & NT2D1	Shared transcription factors (1): YY1
Cells: T47D & Shsy5y	Shared transcription factors (1): GATA3
Cells: T47D & MCF7	Shared transcription factors (1): GATA3
Cells: Gm10847 & Gm18526	Shared transcription factors (1): NFkB
Cells: Gm10847 & Gm19099	Shared transcription factors (1): NFkB
Cells: Gm18526 & Gm19099	Shared transcription factors (1): NFkB
Cells: Shsy5y & MCF7	Shared transcription factors (1): GATA3

Supplementary Table S8

Table S8. The predicted results of Baseline model MLSNet on 165 ChIP-seq datasets.

Datasets	ACC	ROC-AUC	PR-AUC
wgEncodeAwgTfbsBroadDnd41CtcfUniPk	0.931383	0.978935	0.981499
wgEncodeAwgTfbsBroadDnd41Ezh239875UniPk	0.684054	0.763665	0.777782
wgEncodeAwgTfbsBroadGm12878CtcfUniPk	0.919005	0.967925	0.97387
wgEncodeAwgTfbsBroadH1hescCtcfUniPk	0.935915	0.979972	0.981384
wgEncodeAwgTfbsBroadHela3CtcfUniPk	0.898929	0.958678	0.966446
wgEncodeAwgTfbsBroadHela3Ezh239875UniPk	0.669014	0.718492	0.714489
wgEncodeAwgTfbsBroadHela3Pol2bUniPk	0.658577	0.733351	0.726391
wgEncodeAwgTfbsBroadHepg2CtcfUniPk	0.938834	0.979465	0.98238
wgEncodeAwgTfbsBroadHepg2Ezh239875UniPk	0.692131	0.758046	0.731459
wgEncodeAwgTfbsBroadHmecCtcfUniPk	0.948029	0.984691	0.98702
wgEncodeAwgTfbsBroadHsmmCtcfUniPk	0.939727	0.981969	0.983883
wgEncodeAwgTfbsBroadHsmmEzh239875UniPk	0.668831	0.72786	0.691017
wgEncodeAwgTfbsBroadHsmtCtcfUniPk	0.923772	0.971531	0.976061
wgEncodeAwgTfbsBroadHuvecCtcfUniPk	0.934462	0.978041	0.981534
wgEncodeAwgTfbsBroadHuvecEzh239875UniPk	0.723204	0.792442	0.788499
wgEncodeAwgTfbsBroadHuvecPol2bUniPk	0.684741	0.754966	0.739432
wgEncodeAwgTfbsBroadK562CtcfUniPk	0.883263	0.948445	0.958535
wgEncodeAwgTfbsBroadK562Ezh239875UniPk	0.659226	0.692076	0.640563
wgEncodeAwgTfbsBroadK562Hdac2a300705aUniPk	0.690363	0.770552	0.769026
wgEncodeAwgTfbsBroadK562Hdac6a301341aUniPk	0.626966	0.683827	0.696264
wgEncodeAwgTfbsBroadNhaCtcfUniPk	0.93087	0.976827	0.980711
wgEncodeAwgTfbsBroadNhdfadCtcfUniPk	0.933608	0.979407	0.982107
wgEncodeAwgTfbsBroadNhekCtcfUniPk	0.922035	0.971363	0.976117
wgEncodeAwgTfbsBroadNhekPol2bUniPk	0.661804	0.727633	0.731647
wgEncodeAwgTfbsBroadNhlfcCtcfUniPk	0.92403	0.973042	0.977121
wgEncodeAwgTfbsBroadOsteoblCtcfUniPk	0.907048	0.961431	0.968994
wgEncodeAwgTfbsHaibA549Elf1V0422111Etoh02UniPk	0.801366	0.884143	0.895746
wgEncodeAwgTfbsHaibA549Fosl2V0422111Etoh02UniPk	0.882195	0.946655	0.953464
wgEncodeAwgTfbsHaibA549Foxa1V0416102Dex100nmUniPk	0.836762	0.91422	0.917299
wgEncodeAwgTfbsHaibA549GapbV0422111Etoh02UniPk	0.805634	0.881963	0.889299
wgEncodeAwgTfbsHaibA549Sin3ak20V0422111Etoh02UniPk	0.701681	0.766675	0.781758
wgEncodeAwgTfbsHaibA549Taf1V0422111Etoh02UniPk	0.706857	0.775306	0.779464
wgEncodeAwgTfbsHaibEcc1Foxa1sc6553V0416102Dm002p1hUniPk	0.854373	0.920472	0.92928
wgEncodeAwgTfbsHaibGm12878Elf1sc631V0416101UniPk	0.806209	0.891855	0.898718
wgEncodeAwgTfbsHaibGm12878GapbPcr2xUniPk	0.834313	0.910046	0.921586
wgEncodeAwgTfbsHaibGm12878P300Pcr1xUniPk	0.770109	0.850329	0.858236
wgEncodeAwgTfbsHaibGm12878Pax5c20Pcr1xUniPk	0.832246	0.909798	0.914445

wgEncodeAwgTfbsHaibGm12878Pax5n19Pcr1xUniPk	0.808099	0.88815	0.896811
wgEncodeAwgTfbsHaibGm12878Taf1Pcr1xUniPk	0.744128	0.832202	0.835027
wgEncodeAwgTfbsHaibGm12878Yy1sc281Pcr1xUniPk	0.834746	0.915139	0.926646
wgEncodeAwgTfbsHaibGm12891Pax5c20V0416101UniPk	0.807229	0.896443	0.90654
wgEncodeAwgTfbsHaibGm12892Pax5c20V0416101UniPk	0.765963	0.848828	0.861437
wgEncodeAwgTfbsHaibGm12892Taf1V0416102UniPk	0.743097	0.823011	0.836808
wgEncodeAwgTfbsHaibGm12892Yy1V0416101UniPk	0.836537	0.912981	0.922372
wgEncodeAwgTfbsHaibH1hesFcFosl1sc183V0416102UniPk	0.770455	0.843833	0.85889
wgEncodeAwgTfbsHaibH1hesFcGapPcr1xUniPk	0.81802	0.898473	0.908975
wgEncodeAwgTfbsHaibH1hesFcHdac2sc6296V0416102UniPk	0.758374	0.843766	0.847192
wgEncodeAwgTfbsHaibH1hesFcSp1Pcr1xUniPk	0.787043	0.871835	0.884026
wgEncodeAwgTfbsHaibH1hesFcSp2V0422111UniPk	0.834029	0.911317	0.927223
wgEncodeAwgTfbsHaibH1hesFcSrfPcr1xUniPk	0.857214	0.934828	0.944412
wgEncodeAwgTfbsHaibH1hesFcTead4sc101184V0422111UniPk	0.900884	0.960806	0.964175
wgEncodeAwgTfbsHaibH1hesFcYy1sc281V0416102UniPk	0.841936	0.921276	0.928696
wgEncodeAwgTfbsHaibHct116Yy1sc281V0416101UniPk	0.828178	0.908509	0.918338
wgEncodeAwgTfbsHaibHela3GapPcr1xUniPk	0.851753	0.930709	0.940412
wgEncodeAwgTfbsHaibHepg2Cebpbcsc150V0416101UniPk	0.925453	0.975556	0.975155
wgEncodeAwgTfbsHaibHepg2Elf1sc631V0416101UniPk	0.870603	0.939161	0.941903
wgEncodeAwgTfbsHaibHepg2Fosl2V0416101UniPk	0.861794	0.933547	0.940625
wgEncodeAwgTfbsHaibHepg2Foxa1sc101058V0416101UniPk	0.891461	0.95675	0.958348
wgEncodeAwgTfbsHaibHepg2Foxa1sc6553V0416101UniPk	0.896191	0.961496	0.962664
wgEncodeAwgTfbsHaibHepg2GapPcr2xUniPk	0.82943	0.907914	0.913619
wgEncodeAwgTfbsHaibHepg2Hdac2sc6296V0416101UniPk	0.810767	0.893342	0.898611
wgEncodeAwgTfbsHaibHepg2JundPcr1xUniPk	0.842901	0.920085	0.928811
wgEncodeAwgTfbsHaibHepg2P300V0416101UniPk	0.839722	0.920391	0.920613
wgEncodeAwgTfbsHaibHepg2Sin3ak20Pcr1xUniPk	0.741133	0.821614	0.825352
wgEncodeAwgTfbsHaibHepg2Sp1Pcr1xUniPk	0.826992	0.908769	0.915159
wgEncodeAwgTfbsHaibHepg2Sp2V0422111UniPk	0.884393	0.93792	0.951629
wgEncodeAwgTfbsHaibHepg2SrfV0416101UniPk	0.84766	0.917703	0.926138
wgEncodeAwgTfbsHaibHepg2Taf1Pcr2xUniPk	0.741447	0.825202	0.82891
wgEncodeAwgTfbsHaibHepg2Tead4sc101184V0422111UniPk	0.822602	0.899445	0.906996
wgEncodeAwgTfbsHaibHepg2Yy1sc281V0416101UniPk	0.799742	0.884955	0.890597
wgEncodeAwgTfbsHaibK562Cebpbcsc150V0422111UniPk	0.88926	0.952391	0.955761
wgEncodeAwgTfbsHaibK562Elf1sc631V0416102UniPk	0.881209	0.948885	0.950052
wgEncodeAwgTfbsHaibK562Fosl1sc183V0416101UniPk	0.918279	0.96787	0.973419
wgEncodeAwgTfbsHaibK562GapV0416101UniPk	0.8392	0.910302	0.918737
wgEncodeAwgTfbsHaibK562Gata2sc267Pcr1xUniPk	0.879271	0.949102	0.950084
wgEncodeAwgTfbsHaibK562Hdac2sc6296V0416102UniPk	0.76926	0.857183	0.862063
wgEncodeAwgTfbsHaibK562MaxV0416102UniPk	0.808474	0.891237	0.899345
wgEncodeAwgTfbsHaibK562Sin3ak20V0416101UniPk	0.754351	0.835194	0.843336
wgEncodeAwgTfbsHaibK562SrfV0416101UniPk	0.807983	0.893244	0.907457
wgEncodeAwgTfbsHaibK562Taf1V0416101UniPk	0.757255	0.842193	0.849593
wgEncodeAwgTfbsHaibK562Tead4sc101184V0422111UniPk	0.83314	0.911957	0.916261

wgEncodeAwgTfbsHaibK562Yy1V0416101UniPk	0.822633	0.904474	0.909564
wgEncodeAwgTfbsHaibK562Yy1V0416102UniPk	0.833226	0.911497	0.920185
wgEncodeAwgTfbsHaibPanc1Sin3ak20V0416101UniPk	0.716259	0.784473	0.777016
wgEncodeAwgTfbsHaibSknshraP300V0416102UniPk	0.813923	0.894883	0.898154
wgEncodeAwgTfbsHaibSknshraYy1sc281V0416102UniPk	0.809532	0.893332	0.90613
wgEncodeAwgTfbsHaibSknshTaf1V0416101UniPk	0.761957	0.846508	0.848549
wgEncodeAwgTfbsHaibT47dFoxa1sc6553V0416102Dm002p1hUniPk	0.883727	0.952107	0.954791
wgEncodeAwgTfbsHaibT47dGata3sc268V0416102Dm002p1hUniPk	0.843626	0.920193	0.925541
wgEncodeAwgTfbsHaibT47dP300V0416102Dm002p1hUniPk	0.786638	0.86937	0.878348
wgEncodeAwgTfbsSydhA549CebpbIggrabUniPk	0.936854	0.981587	0.98178
wgEncodeAwgTfbsSydhGm10847NfkbTnfalIggrabUniPk	0.792136	0.874644	0.891187
wgEncodeAwgTfbsSydhGm12878CfosUniPk	0.904817	0.95916	0.96366
wgEncodeAwgTfbsSydhGm12878JundUniPk	0.883984	0.947757	0.94863
wgEncodeAwgTfbsSydhGm12878MaxIggmusUniPk	0.751874	0.830228	0.836492
wgEncodeAwgTfbsSydhGm12878NfkbTnfalIggrabUniPk	0.808868	0.887945	0.899284
wgEncodeAwgTfbsSydhGm12878P300bUniPk	0.836701	0.917869	0.921895
wgEncodeAwgTfbsSydhGm12878Stat3IggmusUniPk	0.699647	0.768726	0.768187
wgEncodeAwgTfbsSydhGm12878TbpIggmusUniPk	0.718213	0.797681	0.79992
wgEncodeAwgTfbsSydhGm12891NfkbTnfalIggrabUniPk	0.811423	0.894198	0.904253
wgEncodeAwgTfbsSydhGm12892NfkbTnfalIggrabUniPk	0.807585	0.889827	0.899876
wgEncodeAwgTfbsSydhGm18526NfkbTnfalIggrabUniPk	0.77529	0.868841	0.886885
wgEncodeAwgTfbsSydhGm19099NfkbTnfalIggrabUniPk	0.83727	0.915208	0.922405
wgEncodeAwgTfbsSydhH1hesccCebpbIggrabUniPk	0.940994	0.980968	0.981415
wgEncodeAwgTfbsSydhH1hesccCtbp2UcdUniPk	0.714744	0.786515	0.779844
wgEncodeAwgTfbsSydhH1hesccJundIggrabUniPk	0.850866	0.922618	0.931695
wgEncodeAwgTfbsSydhH1hesccMaxUcdUniPk	0.831625	0.912096	0.923262
wgEncodeAwgTfbsSydhH1hesccSin3anb6001263IggrabUniPk	0.764273	0.843652	0.854328
wgEncodeAwgTfbsSydhHelas3CebpbIggrabUniPk	0.890111	0.955572	0.957035
wgEncodeAwgTfbsSydhHelas3CfosUniPk	0.906181	0.958632	0.962822
wgEncodeAwgTfbsSydhHelas3CjunIggrabUniPk	0.889006	0.948849	0.954519
wgEncodeAwgTfbsSydhHelas3Hae2f1UniPk	0.7994	0.882538	0.881846
wgEncodeAwgTfbsSydhHelas3JundIggrabUniPk	0.900333	0.962108	0.96595
wgEncodeAwgTfbsSydhHelas3MaxIggrabUniPk	0.793214	0.877546	0.883539
wgEncodeAwgTfbsSydhHelas3P300sc584sc584IggrabUniPk	0.843235	0.922123	0.925146
wgEncodeAwgTfbsSydhHelas3Stat3IggrabUniPk	0.806344	0.888229	0.900157
wgEncodeAwgTfbsSydhHepg2CebpbForsklnUniPk	0.906121	0.965568	0.966741
wgEncodeAwgTfbsSydhHepg2CebpbIggrabUniPk	0.944522	0.985682	0.984664
wgEncodeAwgTfbsSydhHepg2CjunIggrabUniPk	0.920758	0.974116	0.977406
wgEncodeAwgTfbsSydhHepg2Corestsc30189IggrabUniPk	0.728041	0.804817	0.812041
wgEncodeAwgTfbsSydhHepg2JundIggrabUniPk	0.937657	0.980928	0.982658
wgEncodeAwgTfbsSydhHepg2MaxIggrabUniPk	0.788453	0.875346	0.883789
wgEncodeAwgTfbsSydhHepg2TbpIggrabUniPk	0.740707	0.816423	0.819975
wgEncodeAwgTfbsSydhHuvecCfosUcdUniPk	0.890412	0.953708	0.958956
wgEncodeAwgTfbsSydhHuvecCjunUniPk	0.9158	0.969384	0.970294

wgEncodeAwgTfbsSydhHuvecGata2UcdUniPk	0.816699	0.894341	0.902744
wgEncodeAwgTfbsSydhHuvecMaxUniPk	0.854843	0.931578	0.938125
wgEncodeAwgTfbsSydhImr90CebpbIggrabUniPk	0.921212	0.974108	0.97376
wgEncodeAwgTfbsSydhK562CebpbIggrabUniPk	0.933528	0.978588	0.978788
wgEncodeAwgTfbsSydhK562CfosUniPk	0.929813	0.976659	0.977674
wgEncodeAwgTfbsSydhK562CjunUniPk	0.911278	0.964107	0.966948
wgEncodeAwgTfbsSydhK562Corestsc30189IggrabUniPk	0.831068	0.90996	0.915279
wgEncodeAwgTfbsSydhK562Gata2UcdUniPk	0.865962	0.932108	0.935383
wgEncodeAwgTfbsSydhK562JundIggrabUniPk	0.874307	0.945385	0.95086
wgEncodeAwgTfbsSydhK562MaxIggrabUniPk	0.810304	0.892472	0.899479
wgEncodeAwgTfbsSydhK562P300IggrabUniPk	0.841496	0.917289	0.923006
wgEncodeAwgTfbsSydhK562TbplgumusUniPk	0.761445	0.842508	0.856971
wgEncodeAwgTfbsSydhK562Yy1UcdUniPk	0.906283	0.961118	0.968112
wgEncodeAwgTfbsSydhMcf10aesCfosTam112hHvdUniPk	0.920601	0.973563	0.972678
wgEncodeAwgTfbsSydhMcf10aesCfosTam14hHvdUniPk	0.914618	0.970603	0.969884
wgEncodeAwgTfbsSydhMcf10aesCfosTamHvdUniPk	0.924682	0.975335	0.975009
wgEncodeAwgTfbsSydhMcf10aesStat3Etoh01bUniPk	0.885815	0.956421	0.958414
wgEncodeAwgTfbsSydhMcf10aesStat3Etoh01cUniPk	0.893417	0.958347	0.959764
wgEncodeAwgTfbsSydhMcf10aesStat3Etoh01UniPk	0.866047	0.941835	0.944056
wgEncodeAwgTfbsSydhMcf10aesStat3Tam112hHvdUniPk	0.892914	0.957136	0.957279
wgEncodeAwgTfbsSydhMcf10aesStat3TamUniPk	0.896604	0.960399	0.961494
wgEncodeAwgTfbsSydhMcf7Gata3UcdUniPk	0.742178	0.813795	0.822749
wgEncodeAwgTfbsSydhMcf7Hae2f1UcdUniPk	0.746092	0.824242	0.822316
wgEncodeAwgTfbsSydhNb4CmycUniPk	0.798022	0.881542	0.892365
wgEncodeAwgTfbsSydhNb4MaxUniPk	0.820647	0.904675	0.913223
wgEncodeAwgTfbsSydhNt2d1Yy1UcdUniPk	0.884306	0.948301	0.959181
wgEncodeAwgTfbsSydhShsy5yGata2UcdUniPk	0.843535	0.919913	0.926432
wgEncodeAwgTfbsSydhShsy5yGata3sc269sc269UcdUniPk	0.809875	0.891084	0.899551
wgEncodeAwgTfbsUchicagoK562EfosUniPk	0.921588	0.970935	0.97494
wgEncodeAwgTfbsUchicagoK562Egata2UniPk	0.776322	0.852574	0.86101
wgEncodeAwgTfbsUchicagoK562Ehdac8UniPk	0.606725	0.66428	0.655445
wgEncodeAwgTfbsUchicagoK562EjundUniPk	0.836924	0.914314	0.923331
wgEncodeAwgTfbsUtaGm12878CmycUniPk	0.717421	0.802359	0.797928
wgEncodeAwgTfbsUtaGm12878CtcfUniPk	0.946727	0.986632	0.987452
wgEncodeAwgTfbsUtaH1hesCmycUniPk	0.589691	0.637882	0.633928
wgEncodeAwgTfbsUtaH1hesCtcfUniPk	0.936335	0.981175	0.981722
wgEncodeAwgTfbsUtaHelas3CtcfUniPk	0.953479	0.988409	0.988162
wgEncodeAwgTfbsUtaHepg2CmycUniPk	0.783352	0.862151	0.865412
wgEncodeAwgTfbsUtaK562CtcfUniPk	0.953276	0.989175	0.989909
wgEncodeAwgTfbsUtaMcf7CtcfUniPk	0.913517	0.968951	0.971436
Average	0.830613	0.8999187	0.903489

Supplementary Table S9

Table S9. The predicted results of TEMNet model on 165 ChIP-seq datasets.

Datasets	ACC	ROC-AUC	PR-AUC
wgEncodeAwgTfbsBroadDnd41CtcfUniPk	0.938244854	0.979842319	0.982335791
wgEncodeAwgTfbsBroadDnd41Ezh239875UniPk	0.681073025	0.722616633	0.746194129
wgEncodeAwgTfbsBroadGm12878CtcfUniPk	0.919811049	0.969662639	0.974733854
wgEncodeAwgTfbsBroadH1hesCtcfUniPk	0.940296193	0.982077631	0.983176469
wgEncodeAwgTfbsBroadHela3CtcfUniPk	0.903420027	0.959800309	0.966564838
wgEncodeAwgTfbsBroadHela3Ezh239875UniPk	0.629577465	0.666507645	0.688002994
wgEncodeAwgTfbsBroadHela3Pol2bUniPk	0.665271967	0.715565877	0.711253279
wgEncodeAwgTfbsBroadHepg2CtcfUniPk	0.943147882	0.981135725	0.983318456
wgEncodeAwgTfbsBroadHepg2Ezh239875UniPk	0.685255921	0.742636039	0.730988837
wgEncodeAwgTfbsBroadHmecCtcfUniPk	0.949906155	0.985958169	0.988300017
wgEncodeAwgTfbsBroadHsmmCtcfUniPk	0.935475709	0.981190542	0.982556603
wgEncodeAwgTfbsBroadHsmmEzh239875UniPk	0.652597403	0.681822496	0.656390764
wgEncodeAwgTfbsBroadHsmtCtcfUniPk	0.926298248	0.973834378	0.977712438
wgEncodeAwgTfbsBroadHuvecCtcfUniPk	0.938042131	0.979479829	0.982917867
wgEncodeAwgTfbsBroadHuvecEzh239875UniPk	0.704946996	0.781104674	0.777258146
wgEncodeAwgTfbsBroadHuvecPol2bUniPk	0.690019194	0.759606915	0.748357153
wgEncodeAwgTfbsBroadK562CtcfUniPk	0.890823178	0.950091544	0.960465874
wgEncodeAwgTfbsBroadK562Ezh239875UniPk	0.626488095	0.643402778	0.647234878
wgEncodeAwgTfbsBroadK562Hdac2a300705aUniPk	0.703244275	0.760920495	0.75393973
wgEncodeAwgTfbsBroadK562Hdac6a301341aUniPk	0.660674157	0.69280055	0.695007795
wgEncodeAwgTfbsBroadNhaCtcfUniPk	0.942134907	0.979063232	0.981840439
wgEncodeAwgTfbsBroadNhdCtcfUniPk	0.939988398	0.981760928	0.983565755
wgEncodeAwgTfbsBroadNhekCtcfUniPk	0.924008992	0.970954965	0.97454672
wgEncodeAwgTfbsBroadNhekPol2bUniPk	0.669761273	0.726115461	0.721226787
wgEncodeAwgTfbsBroadNhlCtcfUniPk	0.92905119	0.973058447	0.976488231
wgEncodeAwgTfbsBroadOsteobCtcfUniPk	0.913408859	0.967749584	0.974038944
wgEncodeAwgTfbsHaibA549Elf1V0422111EtOH02UniPk	0.814726841	0.886791208	0.899212849
wgEncodeAwgTfbsHaibA549Fosl2V0422111EtOH02UniPk	0.891759028	0.956239425	0.962500211
wgEncodeAwgTfbsHaibA549Foxa1V0416102Dex100nmUniPk	0.844061048	0.916477337	0.921560229
wgEncodeAwgTfbsHaibA549GapV0422111EtOH02UniPk	0.814588338	0.886954505	0.899662
wgEncodeAwgTfbsHaibA549Sin3ak20V0422111EtOH02UniPk	0.697478992	0.759389316	0.760396318
wgEncodeAwgTfbsHaibA549Taf1V0422111EtOH02UniPk	0.711700229	0.792604006	0.798883826
wgEncodeAwgTfbsHaibEcc1Foxa1sc6553V0416102Dm002p1hUniPk	0.853511417	0.919626047	0.926584589
wgEncodeAwgTfbsHaibGm12878Elf1sc631V0416101UniPk	0.818202197	0.898459199	0.904723084
wgEncodeAwgTfbsHaibGm12878GapPcr2xUniPk	0.845671759	0.923768236	0.930935823
wgEncodeAwgTfbsHaibGm12878P300Pcr1xUniPk	0.77408143	0.849189091	0.853291204
wgEncodeAwgTfbsHaibGm12878Pax5c20Pcr1xUniPk	0.832245758	0.910788971	0.913293374
wgEncodeAwgTfbsHaibGm12878Pax5n19Pcr1xUniPk	0.810033531	0.891573538	0.899264774
wgEncodeAwgTfbsHaibGm12878Taf1Pcr1xUniPk	0.769948001	0.851413179	0.858930624

wgEncodeAwgTfbsHaibGm12878Yy1sc281Pcr1xUniPk	0.847374543	0.924277509	0.934971554
wgEncodeAwgTfbsHaibGm12891Pax5c20V0416101UniPk	0.781526104	0.866349206	0.876257766
wgEncodeAwgTfbsHaibGm12892Pax5c20V0416101UniPk	0.76154224	0.845505731	0.854979189
wgEncodeAwgTfbsHaibGm12892Taf1V0416102UniPk	0.760422306	0.841000363	0.851884797
wgEncodeAwgTfbsHaibGm12892Yy1V0416101UniPk	0.841705702	0.921612252	0.931035484
wgEncodeAwgTfbsHaibH1hesFcFosl1sc183V0416102UniPk	0.752272727	0.816695602	0.827300771
wgEncodeAwgTfbsHaibH1hesFcGapPcr1xUniPk	0.820695807	0.89675435	0.906234575
wgEncodeAwgTfbsHaibH1hesFcHdac2sc6296V0416102UniPk	0.762840554	0.843671704	0.842844906
wgEncodeAwgTfbsHaibH1hesFcSp1Pcr1xUniPk	0.793978349	0.880811423	0.88921095
wgEncodeAwgTfbsHaibH1hesFcSp2V0422111UniPk	0.838204593	0.908996213	0.92425261
wgEncodeAwgTfbsHaibH1hesFcSrfPcr1xUniPk	0.863205192	0.934748103	0.944334227
wgEncodeAwgTfbsHaibH1hesFcTead4sc101184V0422111UniPk	0.897682162	0.958290521	0.960731624
wgEncodeAwgTfbsHaibH1hesFcYy1sc281V0416102UniPk	0.850027902	0.928450639	0.934750261
wgEncodeAwgTfbsHaibHct116Yy1sc281V0416101UniPk	0.83675913	0.910365466	0.919715015
wgEncodeAwgTfbsHaibHela3GapPcr1xUniPk	0.86242378	0.930268528	0.939332485
wgEncodeAwgTfbsHaibHepg2Cebpbc150V0416101UniPk	0.930787589	0.978947636	0.978739592
wgEncodeAwgTfbsHaibHepg2Elf1sc631V0416101UniPk	0.867777935	0.937968946	0.941470744
wgEncodeAwgTfbsHaibHepg2Fosl2V0416101UniPk	0.866935484	0.937104114	0.943224547
wgEncodeAwgTfbsHaibHepg2Foxa1sc101058V0416101UniPk	0.899928944	0.961409911	0.961920987
wgEncodeAwgTfbsHaibHepg2Foxa1sc6553V0416101UniPk	0.905065816	0.963882601	0.964745418
wgEncodeAwgTfbsHaibHepg2GapPcr2xUniPk	0.829175153	0.904262725	0.910500008
wgEncodeAwgTfbsHaibHepg2Hdac2sc6296V0416101UniPk	0.817195661	0.897438227	0.902667737
wgEncodeAwgTfbsHaibHepg2JundPcr1xUniPk	0.848172446	0.927091166	0.934683421
wgEncodeAwgTfbsHaibHepg2P300V0416101UniPk	0.854450645	0.930327681	0.931638099
wgEncodeAwgTfbsHaibHepg2Sin3ak20Pcr1xUniPk	0.761860894	0.839849356	0.846064579
wgEncodeAwgTfbsHaibHepg2Sp1Pcr1xUniPk	0.84020825	0.917129821	0.923991026
wgEncodeAwgTfbsHaibHepg2Sp2V0422111UniPk	0.88150289	0.937069222	0.94706665
wgEncodeAwgTfbsHaibHepg2SrfV0416101UniPk	0.842406877	0.915098274	0.923762458
wgEncodeAwgTfbsHaibHepg2Taf1Pcr2xUniPk	0.754374901	0.837108994	0.839211399
wgEncodeAwgTfbsHaibHepg2Tead4sc101184V0422111UniPk	0.833836351	0.909497626	0.915469472
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wgEncodeAwgTfbsHaibSknsbraP300V0416102UniPk	0.817972855	0.899633282	0.903155547
wgEncodeAwgTfbsHaibSknsbraYy1sc281V0416102UniPk	0.823616115	0.902178518	0.910701885
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wgEncodeAwgTfbsUtaHepg2CmycUniPk	0.789931585	0.857649912	0.861308417
wgEncodeAwgTfbsUtaK562CtcfUniPk	0.955994839	0.988598867	0.989214392
wgEncodeAwgTfbsUtaMcf7CtcfUniPk	0.917158975	0.968807619	0.970049539
Average	0.837094712	0.902132217	0.906037431